PN搭建GCN框架 方案一

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Technical paper

Petri-net-based deep reinforcement learning for real-time scheduling of automated manufacturing systems

Jiliang Luo [a,c,*], Sijia Yi [b,c], Zexuan Lin [a,c], Hongbin Zhang [a,c], Jiazhong Zhou [a,c]

[a] College of Information Science and Engineering, Huaqiao University, Xiamen 361021, China
[b] College of Mechanical Engineering and Automation, Huaqiao University, Xiamen 361021, China
[c] Fujian Engineering Research Center of Motor Control and System Optimal Schedule, Xiamen 361021, China

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ABSTRACT

This work focuses on real-time and joint optimizing issues to handle the routing of automated guided vehicles and allocating processing tasks for automated manufacturing systems. A real-time scheduling approach is proposed based on Petri nets and deep reinforcement learning. First, the considered system is modeled with a placed-timed Petri net, which plays the role of the environment for deep reinforcement learning. Second, a novel graph convolutional network (GCN), called Petri-net-GCN, is designed by using the topological structures of a Petri net model such that it can adapt to changes in Petri-net structures caused by abrupt events such as new orders and failures of devices. Third, a deep Q-network method is presented to train Petri-net-GCN to predict the minimal time to lead a Petri net to its goal if an action (transition) is taken at a present state. Consequently, Petri-net-GCN can replace scheduling rules to schedule an automated manufacturing system in real time. Finally, numerical experiments are carried out where there exist failures of devices and new orders. The results show that Petri-net-GCN can adapt to the Petri-net changes caused by abrupt events and performs much better than the first-in-first-out and longest-processing-time-first rules, which are applicable for real-time scheduling issues in practice.

1. Introduction

As Industry 4.0 and intelligent manufacturing continuously evolve [1], it becomes imperative to enhance the degrees of flexibility and efficiency in automated manufacturing systems (AMSs). They are required to process multiple parts simultaneously on various equipment and adjust the production process and quantity of parts according to the dynamically changing demand [2]. Due to the customization and uncertainty of market [3–5], their dynamics become more complex. Their state space grows explosively due to robotic devices such as automated guided vehicles (AGVs) and computerized numerical control cells.

As a central topic in the manufacturing field, scheduling AMSs is a classical combinatorial optimization problem [6–9]. To effectively address such problems, it is necessary to formally construct an information-physical system. Its information space should not only comprehensively extract relevant scheduling data but also be compact and user-friendly. Petri nets can describe sequential, parallel, synchronous, and asynchronous behaviors of AMSs and other discrete event systems. Furthermore, they have the capability to capture changes in the system's state. What is even more important is that the scalability of Petri nets grows linearly with the number of units in the system, providing extremely high modeling efficiency, making them widely used in the development of scheduling methods [10–14]. The state-of-the-art scheduling methods based on Petri nets fall into three categories: metaheuristic, tree search, and machine learning.

Metaheuristic algorithms are developed based on Petri nets and local search. Xing et al. [15] propose a genetic algorithm to minimize the makespan of AMS, where a chromosome represents a permutation of operations coded by an optimal deadlock avoidance policy of place-timed Petri nets. Luo et al. [16] develop a genetic algorithm to schedule a flexible assembly system, where a redefined deadlock prevention policy based on Petri nets, is embedded, and a perturbation strategy is used to escape from local optima. Han et al. [17] propose a hybrid particle swarm optimizer (PSO) to schedule AMSs, where operation routes are encoded as particles based on timed Petri nets, and a deadlock controller is used to decode deadlock-prone particles into feasible schedules. PSO is further extended by Li et al. [18] such that it can be applied to flexible assembly systems without intermediate

* Corresponding author at: College of Information Science and Engineering, Huaqiao University, Xiamen 361021, China.
E-mail addresses: jlluo@hqu.edu.cn (J. Luo), yisijia@stu.hqu.edu.cn (S. Yi), linzexuan@stu.hqu.edu.cn (Z. Lin), hopingzhang@foxmail.com (H. Zhang), zhoujiazhong@hqu.edu.cn (J. Zhou).

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30-GCN转.pdf

基于这篇论文搭建GCN框架

输入

只用place特征矩阵X_p

对每个place p 设置

$$X_p = [m(p), m_g(p), v(p), d(p)]$$

分别为

- m(p)：当前p中的token分布
- m_g(p)：目标状态p的token分布
- v(p)：当前token再该place已停留的时间
- d(p)：该place的延迟时间（加工时间）

此处暂时先不设置T的初始特征

图结构

用已有的

- T的前置库所：Pre-incidence matrix
- T的后置库所：Post-incidence matrix

网络

按照论文的给定4类slice

输入矩阵：

$$X_p^{(0)} \in \mathbb{R}^{|P| \times 4}$$

P2P：把4维原始状态映射到高维

$$X^{j+1} = X^j \cdot W_p + B_p$$

其中

$$W_p \in \mathbb{R}^{4 \times \lambda_p}$$

$$B_p \in \mathbb{R}^{|p| \times \lambda_p}$$

$$H_p^{(1)} = \sigma(X_p^{(0)} W_p + b_p)$$

（可以尝试ReLU函数）

4：初始特征维度

$|P|$：Petri网中place的数量

λ_p：超参数设置（4维特征少，升维参数）

P2T：

$$X^{j+1} = (C^-)^\top X^j W_{p\to t} + (C^+)^\top X^j W_{p\leftarrow t}$$

其中：

$X^j \in \mathbb{R}^{|P|\times\lambda_p}$

$X^{j+1} \in \mathbb{R}^{|T|\times\lambda_t}$

$W_{p\to t}, W_{p\leftarrow t} \in \mathbb{R}^{\lambda_p\times\lambda_t}$

$$H_t^{(2)} = \sigma((C^-)^\top H_p^{(1)} W_{p\to t} + (C^+)^\top H_p^{(1)} W_{p\leftarrow t}$$

T2P:

$$X^{j+1} = (C^+)^\top X^j W_{t\to p} + (C^-)^\top X^j W_{t\leftarrow p}$$

其中：

$X^j \in \mathbb{R}^{|T|\times\lambda_t}$

$X^{j+1} \in \mathbb{R}^{|P|\times\lambda_p}$

$W_{t\to p}, W_{t\leftarrow p} \in \mathbb{R}^{\lambda_t\times\lambda_p}$

$$H_t^{(3)} = \sigma((C^+)^\top H_t^{(2)} W_{t\to p} + (C^-)^\top H_t^{(2)} W_{t\leftarrow p}$$

（还需要一轮/多轮P2T供后续T2T使用）

如：

$$H_t^{(4)} = \sigma((C^-)^\top H_p^{(3)} W_{p\to t} + (C^+)^\top H_p^{(3)} W_{p\leftarrow t}$$

T2T:

$$X^{j+1} = X^j \cdot W_t + B_t$$

其中

$$W_t \in \mathbb{R}^{\lambda_t\times 1}$$

$$B_p \in \mathbb{R}^{|T|\times 1}$$

T：Petri网中变迁的数量

$$q = H_p^{(4)} W_{out} + b_{out}$$

其中：

$q \in \mathbb{R}^{|T|}$

输出

每个q对应一个变迁的分数

决策

先做action mask，再从安全动作里取出最大值

输出的格式应当是

$$q = [q_1, q_2, \ldots, q_{|T|}]$$

动作掩码

拆分成三部分（不能确定是否可行，约束过强）

$$mask(t) = enabled(t) \wedge resource(t) \wedge time(t)$$

enabled：来自Petri网定义，条件是前置库所有足够的token并满足弧值

resource：资源约束，机械手是否空闲/腔室是否空闲

time：时间是否违反驻留时间约束

大致给出了一些想法：

- t_{now} 作为当前时间
- D_{time} 作为该token的deadline时间
- W_{time} 表示进入某库所加工腔之后的消耗的时间

那么有：

$$time - safe(t) = 1[t_{now} + \Delta t(t) \leq D]$$